

FLEXHAND · REQUIREMENTS

FlexHand requirements

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DOCUMENT CONTROL

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Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/flexhand		
Source	docs/03-requirements.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.5	2026-10-01	Amish Chadha	Status from FXH-CAL-001 v0.3 for the constructable design (FXH-DDR-003); R12 reported against the value-engineering target
0.4	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002); R2 reworded to mild tone, R5 lower limit 4 s at design load, status from FXH-CAL-001 v0.2
0.3	2026-09-25	Amish Chadha	TRL 3 status from FXH-CAL-001; decisions of FXH-DDR-001 applied (design load kept, R7 target kept pending therapist review, 20 mm cuff limit)
0.2	2026-09-25	Amish Chadha	First measurable requirements for TRL 2, with concept status
0.1	2026-09-24	Amish Chadha	Initial scaffold

FlexHand requirements

These requirements are checked by calculation in FXH-CAL-001 v0.3, for the constructable design of FXH-DDR-003. Seven are met on paper, two are at risk (R2 and R3), one is not met (R7, forearm pack mass) and three cannot be verified until TRL 4. On 2026-09-25 Amish accepted the recommendations raised by the TRL 3 calculations (FXH-DDR-002): R2 now names mild flexor tone (N1), and the lower end of the R5 stroke-time range is 4 s at design load (N5). The fingertip thimble (N2), the balance pulleys (N3) and the lighter couplings and perforated cuff (N4) are in the design. The 450 g pack target stays until a therapist has reviewed it, which waits on the choice of a clinical co-design partner (FXH-DDR-001, D1 and O1).

Table 1. Requirements and TRL 3 status.

ID	Requirement	Target	Verification	TRL 3 status (FXH-CAL-001)
R1	Move the four fingers through a useful range	MCP 0 to 70°, PIP 0 to 90°, within therapist-set limits	Tendon excursion calculation; later bench test on a finger model	Met: 24.8 mm excursion needed, 29.8 mm stroke
R2	Extend fingers against mild flexor tone (MAS 1 to 1+) (N1)	30 N extensor tendon force per finger at the finger	Torque and friction calculation; published stiffness data	At risk: 30 N covers the 18 N a MAS 1+ finger needs; 0.492 N·m peak at the spool, 125 % of the gearbox's recommended continuous load (RMS 68 %, peak 63 % of intermittent)
R3	Limit force on the hand	Software limit 40 N per finger from motor current; mechanical breakaway at 60 N or less per tendon	Calculation; later bench test	At risk: the balance pulleys (N3) keep both fingers of a pair at equal tension, so the pair limit holds each finger to 40 N (was up to 80 N); friction still spreads the force from 27 to 47 N at a 40 N setting
R4	Deliver a high repetition dose	300 or more full flexion and extension cycles in a 60 min session	Speed calculation	Met: 445 cycles per hour at full speed; 360 at the 4 s lower stroke limit; holds for strokes up to 5 s
R5	Move slowly and smoothly	Stroke time adjustable from 4 to 15 s at design load (N5), down to 3 s at lighter loads; joint speed 30 °/s or less; dwell at each end adjustable	Speed calculation; firmware sketch	Met: fastest stroke at design load 3.3 s (3.9 s with pessimistic friction; 2.6 s unloaded)
R6	Run several sessions per charge	3 or more 60 min sessions at design load	Power budget	Met: 3.8 sessions at 3.80 Wh each
R7	Light enough to wear seated	Hand-side parts 120 g or less; forearm pack 450 g or less (pack target to be revisited with a therapist, D1)	Mass estimate, then weighing	Not met: hand 117 g (met); pack 723 g in the constructable design (639 g in the concept, 675 g before N4)
R8	Quick to put on and take off	A care partner dons it in 5 min or less and removes it in 1 min or less	Design review with users	Not verifiable at TRL 3; at risk on flexed, spastic fingers
R9	Stop and release on demand	Physical stop cuts motor power within 100 ms; tendons released by hand in 10 s or less without tools	Calculation; later bench test	Not verifiable at TRL 3: stop in about 11 ms by calculation; release time needs a bench test
R10	Fit most adult hands	Hand length 170 to 205 mm with three glove sizes and adjustable cuffs	Anthropometric check	Met on paper: sizes scaled 0.95, 1.00 and 1.15 from the model
R11	Keep contact pressure tolerable	Mean pressure under a finger cuff or thimble 50 kPa or less at design load (proposed)	Contact area calculation	Met on paper: 23 to 40 kPa on the medium hand, 46 kPa worst on a small hand's little finger, with the fingertip thimble sharing the load (N2; was 46 to

ID	Requirement	Target	Verification	TRL 3 status (FXH-CAL-001)
		limit, to be set with a therapist)		84 kPa); load sharing assumed, to check at TRL 4
R12	Low cost and buildable	Parts cost against the USD 500 value-engineering target (a control target, not a limit); no custom PCB for the first build	Priced BOM	Estimated cost USD 303.90, USD 196.10 under the target; no custom PCB
R13	Record what was done	Device logs session time, cycle count and peak motor current; therapist limits cannot be changed from the user controls	Design review	Not verifiable at TRL 3: storage ample (1 MB holds decades of summaries)

■ Assumptions

- Design load: 30 N of extensor tendon force per finger at full extension (D5). Checked against the finger stiffness measured by [Heung et al., *Frontiers in Bioengineering and Biotechnology*, 2020](#): it covers a finger with MAS 1+ tone with a margin of about 1.7, not a finger with MAS 3 tone. It is consistent in scale with the Columbia orthosis, whose single motor provides about 100 N peak for four fingers ([Park et al., *arXiv 1802.06131*](#)).
- Tendon excursion for 70° at the MCP and 90° at the PIP is 24.8 mm, taking dorsal moment arms of 10 mm and 8 mm; the stroke adds 5 mm for cuff compliance.
- Transmission efficiency is 0.659: capstan friction with a coefficient of 0.1 over 180° of sheath bend, one idler at 95 % and one balance pulley at 95 %.
- The extensor load of each finger is shared by the middle-phalanx cuff and the fingertip thimble in proportion to their contact areas.
- The forearm rests on a support during sessions. The device is not designed to be worn while walking.
- Sessions are supervised by a care partner, at least until a therapist agrees otherwise.

Safety: R3, R9 and R11 are safety requirements. A design that misses any of them must not be worn by a person, even at TRL 4 bench stage. R11 is met only on paper and R3 is at risk, so nothing may be worn on the strength of these numbers.